

[DC] Developing Virtual Reality Rehabilitation Tools for PTSD

Ilan Vol*
Department of Biomedical
Engineering
Ben Gurion University

ABSTRACT

The world is full of conflicts which cause widespread psychological trauma, leading to conditions such as post-traumatic stress disorder (PTSD). While traditional therapies offer some relief, the global mental health crisis, limited access to care, technical limits, and lack of personalization hinder effective treatment. Virtual Reality (VR) is emerging as a promising tool to address these challenges. Building upon existing virtual exposure therapy tools like BRAVEMIND, I will adapt and expand its capabilities to align with the specific needs of civilians affected by the Israeli-Palestinian conflict. Additionally, I will develop a novel VR tool that focuses on Place Attachment theory, facilitating emotional processing. Ultimately, this research seeks to contribute to the development of accessible and effective mental health interventions, particularly for individuals who have experienced trauma and loss, as part of my wider thesis in which I am developing extended reality tools for rehabilitation.

Index Terms: Virtual Reality, Rehabilitation, mental rehabilitation, PTSD, VRET, place attachment

1 INTRODUCTION

Recent conflicts like the Israeli-Palestinian conflict - which resulted in tragic loss of life, widespread injuries, and kidnapping of civilians, leave survivors with a range of mental health issues including anxiety and depression, and if trauma related symptoms persist they may indicate post-traumatic stress disorder (PTSD) [1]. These experiences can severely hinder daily functioning and relationships and increase the risk of suicide and self-harm.

This crisis is compounded by several challenges. A lack of rehabilitation facilities, funding and healthcare professionals [2], Stigma surrounding mental health treatment [3], and high rates of treatment attrition [4], constitute an obstacle to providing, seeking, and receiving mental health help. Furthermore, traditional one-size-fits-all therapeutic methods may not fully address the complex needs of individuals suffering from trauma, which are crucial to tailoring treatment plans and ensuring optimal outcomes [5]. As a result, many individuals with psychological injuries struggle to access the care they require, leading to prolonged suffering, diminished quality of life, and difficulty to recover and return to some kind of routine.

In my work so far, I have focused on the use of virtual reality tools for physical rehabilitation, such as post-stroke and orthopedic rehabilitation tools to assess and train obstacle avoidance. In the second half of my Ph.D., highlighted in this document, I am turning my focus to these mental health challenges.

Virtual Reality (VR), in which users enter a computer-generated environment that they can explore and interact with, has emerged as a promising tool to address these challenges. VR offers several advantages, such as the ability to create flexible and naturalistic tasks that can be controlled while avoiding the logistical challenges and high costs associated with traditional treatment. In addition,

VR provides a unique opportunity to document and evaluate patient progress by tracking and recording data from them combined with additional mobile equipment. These quantitative data facilitate evidence-based practices, ultimately leading to more targeted and personalized rehabilitation strategies.

In fact, the use of VR to treat PTSD is already well established [6]. VR exposure therapy (VRET) offers a safe and controlled environment to gradually confront trauma-related triggers. By repeatedly exposing patients to feared stimuli, VRET helps reduce PTSD symptoms and promote healing. It can also improve repeated exposures to the story of a death, induce a sense of closeness to the deceased, and address unresolved issues. However, such tools must be updated and tailored to the specific needs of the patient, taking into account their experience while developing the virtual environment.

Going beyond exposure therapy, VR also offers the potential for other treatment approaches such as attachment theory [7]. This theory suggests that secure attachments foster safety and healthy development, but when an attachment figure is lost, separation anxiety triggers behaviors to re-establish the connection [8,9].

The two-track model of bereavement offers a comprehensive framework to understand grief and cope with loss [10]. Track I focuses on adjusting to life without the lost figure, involving detachment to reduce pain and facilitate adaptation. Track II emphasizes integrating memories and experiences of the lost figure into the present. These tracks are complementary.

Traditional VRET, focused on passively re-experiencing traumatic memories, is insufficient as not all patients respond to it. A dual-track approach that combines elements of both tracks to create a balanced therapeutic experience is needed. This approach mitigates distress, fosters emotional processing, and promotes healthy responses to trauma. By addressing past losses and current realities, it supports healing and adjustment.

After developing the skills needed and creating rehabilitation tools with our clinical collaborators in the Departments of Physical Therapy and Occupational Therapy during my M.Sc., I will modify existing VRET tools to align with our local cultural, social, and environmental contexts, and develop an innovative VR tool utilizing place attachment dual-track exposure, and conduct thorough usability and feasibility as my doctoral thesis. This research will address two primary questions: (1) How effective is culturally tailored VRET in reducing PTSD symptoms compared to traditional methods? and (2) Can incorporating place attachment theory in VR tools enhance emotional processing in trauma patients?

2 RESEARCH PLAN

My current work focuses on: customizing VR environments to local cultural contexts to improve patient engagement and therapy retention; Integrating dual-track bereavement frameworks enhancing emotional adjustment in trauma patients. To do so, I will use the Blender graphics engine to create assets and environments and the Unity game engine to create the XR tasks. I aim for cross-platform VR headset compatibility, with emphasis on the Meta Quest ecosystem. Our work is aimed at human participants, both healthy and facing rehabilitation challenges which require ethics approval - the basic usability tests are already IRB approved, and Helsinki approval for patients is underway. Participants will be split into intervention and

*e-mail: vol@post.bgu.ac.il

waiting list control groups. Pre- and post-intervention assessments will include homesickness (Homesickness Scale), PTSD (PCL-5), anxiety (GAD-7), depression (PHQ-9) and secure attachment (State Adult Attachment Measure). We expect improvements in the intervention group compared to the control and will also examine moderating factors such as attachment style (ECR-12) and community resilience (CCRAM-10). Post-session VR engagement and emotional responses (PANAS) will also be evaluated.

2.1 BRAVEMIND - Exposure Therapy

The main VRET tool currently in use is BRAVEMIND. This software, developed by Rizzo et al. [6], is a verified clinical tool to assess and treat PTSD currently deployed in the US. It enables clinicians to customize and control immersive scenarios, allowing gradual exposure to environments that mirror their traumatic experiences. These environments and scenarios are very culture and situation dependent, requiring significant adaptation with clinical supervision. This tool is already being adapted for the Russia-Ukraine conflict [6] and I aim to adapt it to the Israeli-Palestinian conflict.

My first stage has been to learn and master the existing tool and its underlying codebase, which is now set up in our lab, and preparing a demo station. I then gathered a group of professional local clinicians for a focus group session. These clinicians represented a wide variety of clinical practices (e.g. psychologist, psychiatrist, social worker) and all have practical experience treating PTSD caused by the current conflict. This focus group was also followed by a detailed structured survey with the goal of mapping out the most critical adaptations required for local deployment of the system. These included mapping out the different scenarios (e.g. trapped in a safe room, escaping while leaving friends behind, etc.), different local populations, different graphics alterations (e.g. building types, local insignia, localized graffiti, versions of arabic etc.) and different interactive events to be built into the scenarios. I am now performing these adaptations in an iterative design process with our clinical collaborators. While the main goal of this line of research is translational, I am also testing a series of new features which are not included in the original BRAVEMIND such as shared clinician-patient virtual spaces, the integration of photorealistic scenes, and more with an eye to both improving clinical outcome and to gaining a deeper basic science understanding of the brain and biological factors underlying PTSD.

2.2 PAVR - Place Attachment

While the VRET system shows great results in treating PTSD, it still does not cover all of the cases raising the need for additional tools. One potential direction is going beyond passive exposure to active interaction. Therefore, in parallel to the aforementioned work I am building our own tool, re-PAVeR which utilizes clinical protocols built upon place attachment and bereavement theories.

In this tool, the user is placed into a ruined environment. They must then physically engage, interact, and fix the simulated environment under the guidance of the therapist that uses gradual exposure therapy techniques, adjusting the intensity based on the patient's distress levels. E.g., a patient may start in a safe room with an upturned sofa, crooked pictures, bullet holes, blood stains etc. They are then guided to correct each feature and restore the environment, fostering a sense of safety and attachment. Positive actions are reinforced with sensory enhancements (e.g., audio). The therapy also includes future-oriented scenarios, helping patients envision and choose their desired future. For example, I implemented a virtual safe-room which includes unique local features (e.g. a Noah's Ark filtration system) and furniture which the patient can return to place, bullet holes and blood stains they can repair, etc. My next steps are to complete this scenario and integrate it into practical clinical use while developing additional scenarios and the ability to customize and personalize treatment scenarios to the patient's specific needs.

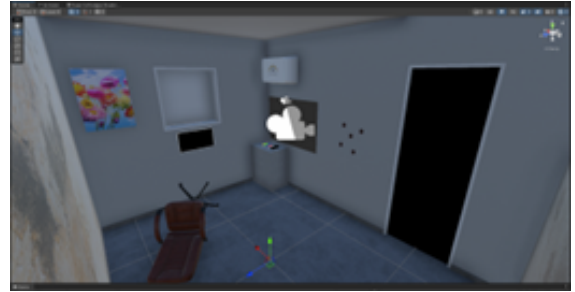


Figure 1: Screenshot of my VR place attachment tool - a residential protected space with destroyed walls, blood stains, bullet holes, overturned furniture, a picture, and a drawing board with markers.

3 DISCUSSION

Thus, to optimize VR rehabilitation I am interested in learning from the experience of others. Some specific questions of interest are:

- What are the best practices for bringing VR from the lab to practical clinical uses for mental and physical challenges?
- When creating a VR therapeutic environment, should you emphasize realism or simplicity?
- How can we best enable clinical staff to support treatment? Should they enter the virtual space or control it from without?

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